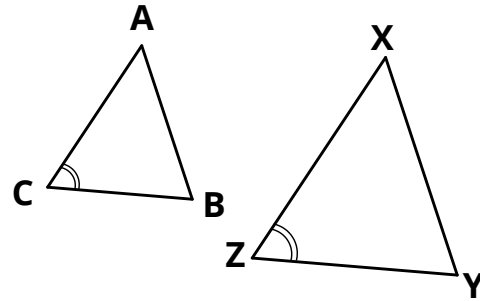


Proving triangles similar



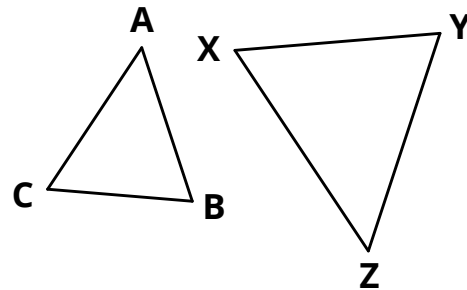
- 1 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle XYZ$.



To prove: $\triangle ABC \sim \triangle XYZ$

Statements	Reasons
1. $\frac{AC}{XZ} = \frac{BC}{YZ}$	Given
2. $\angle ACB \cong \angle XZY$	Given
3. $\triangle ABC \sim \triangle XYZ$	

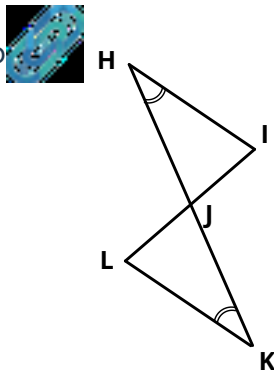
- 2 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

Statements	Reasons
1. $\frac{AC}{YZ} = \frac{AB}{XZ} = \frac{CB}{XY}$	Given
2.	

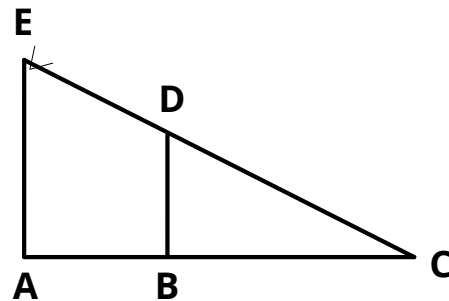
- 3 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

Statements	Reasons
1. $\angle IHJ \cong \angle LKJ$	Given
2. $\angle IJH \cong \angle LJK$	Vertical angles congruence theorem
3. $\parallel$	$\parallel$

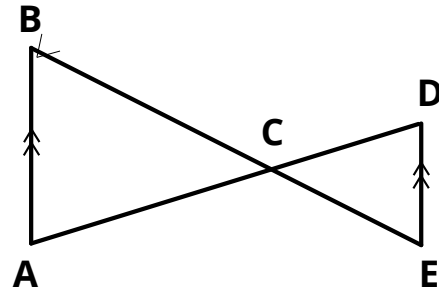
- 4 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

Statements	Reasons
1. $\frac{AC}{BC} = \frac{EC}{DC}$	Given
2. $\angle ACE \cong \angle BCD$	Reflexive property of congruence
3. $\parallel$	$\parallel$

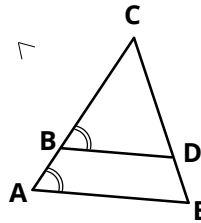
- 5 Use the information given in the diagram to complete the two-column proof that shows that the two triangles are similar.



To prove: The two triangles are similar.

Statements	Reasons
1. $\overline{AB} \parallel \overline{DE}$	Given
2. $\angle BAC \cong \angle EDC$	Alternate interior angles theorem
3. $\angle DCE \cong \angle ACB$	Vertical angles congruence theorem
4. $\parallel$	$\parallel$

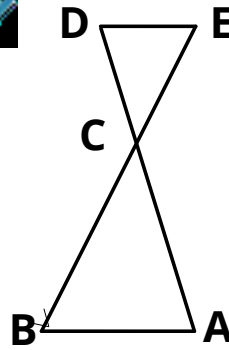
- 6 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ACE \sim \triangle BCD$.



To prove: $\triangle ACE \sim \triangle BCD$

Statements	Reasons
1. $\angle CBD \cong \angle CAE$	Given
2. $\parallel$	$\parallel$
3. $\triangle ACE \sim \triangle BCD$	Angle-angle similarity (AA $\sim$)

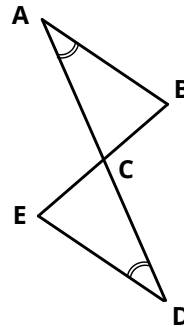
- 7 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle DEC$.



To prove: $\triangle ABC \sim \triangle DEC$

Statements	Reasons
1. $\frac{BC}{CE} = \frac{AC}{CD}$	Given
2. $\parallel$	$\parallel$
3. $\triangle ABC \sim \triangle DEC$	Side-angle-side similarity (SAS $\sim$)

- 8 Use the information given in the diagram to complete the two-column proof that shows that $\triangle ABC \sim \triangle DEC$.



To prove: $\triangle ABC \sim \triangle DEC$

Statements	Reasons
1. $\angle BAC \cong \angle EDC$	Given
2. $\parallel$	$\parallel$
3. $\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)



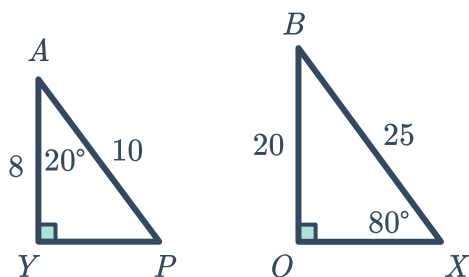
Additional questions

9 Determine whether each statement is true or false:

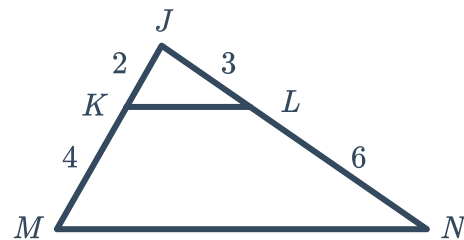
- a If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar.
- b If two sides of one triangle are proportional to two corresponding sides of another triangle, then the triangles are similar.
- c If two sides of one triangle are congruent to two corresponding sides of another triangle, then the triangles are similar.
- d If two triangles are similar, then their corresponding sides are congruent.
- e If two triangles are similar, then their corresponding angles are congruent.

10 Determine whether each pair of triangles is similar. If so, write a similarity statement and justify with a similarity postulate or theorem. If not, explain why not.

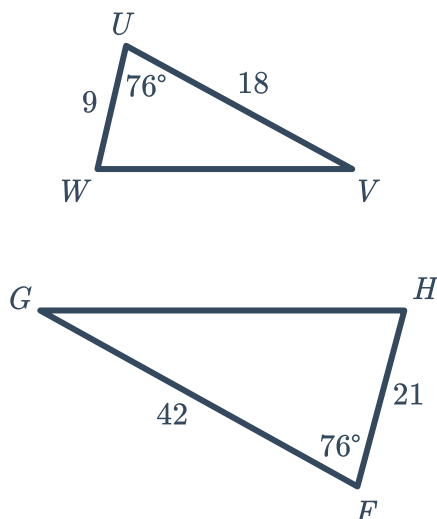
a



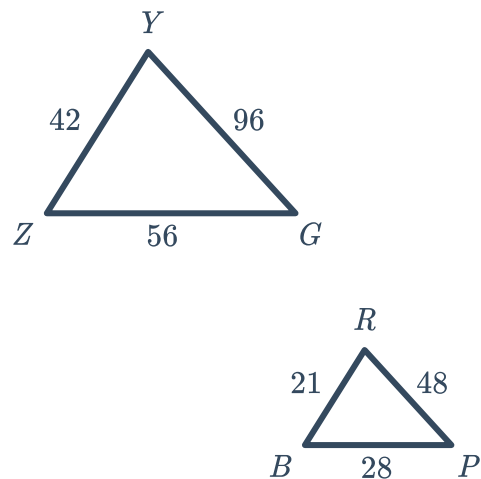
b



c

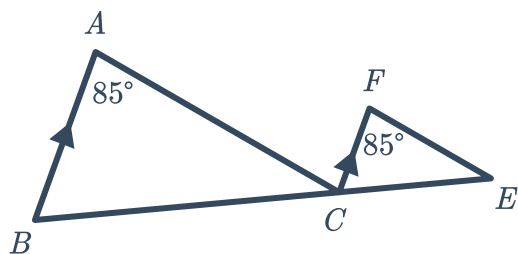


d

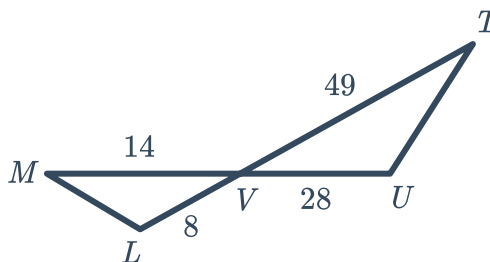


- 11 Show that each of the following pairs of triangles are similar by identifying the similarity postulate or theorem and listing the pairs of sides and angles that justify it.

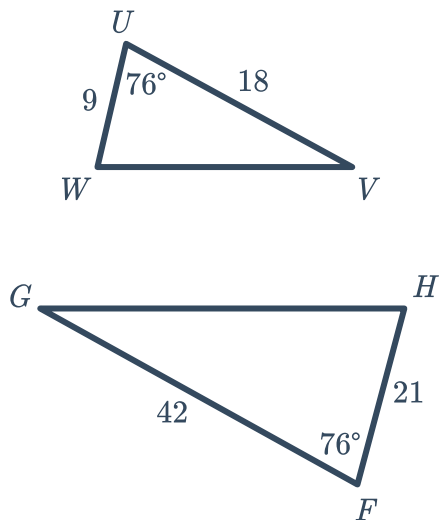
a



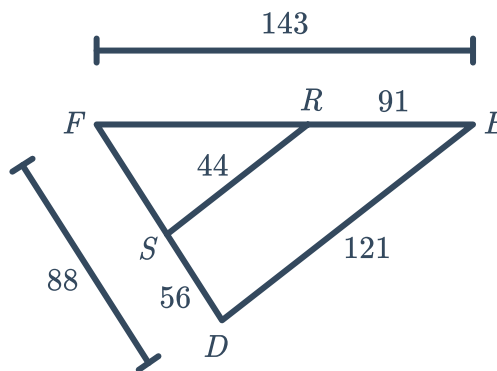
b



c



d



- 12 Given: $\angle S \cong \angle V$
Determine the value of x that makes $\triangle VUW \sim \triangle SUT$.

